NAME: _____ NetID:_____

INFO 4220 – Networks II: Market Design Spring 2022 – Prelim 1

Problem 1 (10 points, 2 points for each question)

1.1. In a one-to-one matching problem where we are matching musicians to singers. Define what is the musician optimal matching and explain how you can find it.

The musician optimal matching is the most preferred stable matching for musicians. In order to find it we can use the deferred acceptance algorithm with musicians proposing.

1.2 For each of the following games:

a) Identify all possible equilibrium(s) in dominant actions
b) Identify all possible Nash Equilibrium(s)
c) Check if any of the equilibrium(s) you identified in a) or b) is/are pareto dominated by other outcomes. If so, why isn't(aren't) the pareto superior outcome(s) an equilibrium?

		Player B		
		Left	Middle	Right
Player A	Up	(2,2)	(2,2)	(-10,10)
	Middle	(2,2)	(5,5)	(-8,10)
	Down	(10,-10)	(10,-8)	(-5,-5)

Note that the best response of player A to whatever action of player B is to play down: $BR^A(Left) = Down, BR^A(Middle) = Down,\ BR^A(Right) = Down$

And the best response of player B to whatever action of player A is to play right: $BR^B(Up) = Right, BR^B(Middle) = Right,\ BR^B(Down) = Right$

This means that there is one equilibrium in dominant actions (Down, Right), which is also a Nash Equilibrium.

(Down, Right) is pareto dominated by (Up, Left), (Up, Middle), (Middle, Left), and (Middle, Middle). While those options would actually leave both players better off, they cannot be a N.E. because in each of them a player has an incentive to change their action.

1.3 Explain why it is important to use a mechanism that produces stable matchings in the national medial residence program

If the program uses a mechanism that is not stable, over time the market will tend to fail. In most countries/places where unstable mechanisms were used, the markets have failed over time. This happens because if the mechanism is not stable, there will be blocking pairs and hospitals and doctors in those pairs would prefer to not follow through with the match prescribed by the mechanism and match out of the system.

1.4 In a one-to-one matching problem we have 3 musicians (m_i) and 3 singers (s_i) with the following preferences:

P_{m_1}	P_{m_2}	P_{m_3}	P_{s_1}	P_{s_2}	P_{s_3}
s_1	s_3	s_2	m_1	m_1	m_2
s_2	s_2	s_1	m_2	m_2	m_3
s_3	s_1	s_3	m_3	m_3	m_1

Prove if the matching μ given by $\mu(m_1) = s_1, \mu(m_2) = s_2, \mu(m_3) = s_3$ is stable (or not)

We need 2 conditions: Individual rationality and absence of blocking pairs.

Individual rationality is trivial to prove, as no doctor finds any hospital unacceptable, and no hospital finds any doctor unacceptable. Thus, it is impossible that any doctor/hospital would prefer to be unmatched.

There is a blocking pair: (m_2, s_3) because they are not matched together, and m_2 would prefer s_3 to s_2, and s_3 would prefer m_2 to m_3

Thus, this matching is not stable

1.5 What does it mean that in the school choice problem (with strict priorities) efficiency and stability are incompatible? What algorithm did the Boston Public School system choose to use in the redesign of their program? Is this algorithm efficient? Is it strategyproof?

It means that in general you can achieve either efficiency (with the top trading cycle algorithm) or stability (with the deferred acceptable algorithm). Moreover, whenever efficiency and stability can be achieved at the same time, there is no algorithm that will always find that solution whenever it exists. The BPS was using immediate acceptance and switched to deferred acceptance (with students proposing). Deferred acceptance is not efficient, and it is strategyproof (for students, but note that schools in theory only have priorities, so they have no reason to misrepresent/manipulate priorities).

Problem 2 (20 points)

Consider 4 students i_1, i_2, i_3, i_4 and 4 schools s_1, s_2, s_3, s_4. Each school has a capacity of 1. True students' preferences and schools' priorities are given by the following tables:

P_{i_1}	P_{i_2}	P_{i_3}	P_{i_4}
s_2	s_2	s_2	s_2
s_4	s_1	s_3	s_3
s_3	s_3	s_4	s_4
s_1	s_4	s_1	s_1

P_{s_1}	P_{s_2}	P_{s_3}	P_{s_4}
i_1	i_1	i_4	i_4
i_2	i_2	i_3	i_3
i_3	i_3	i_2	i_2
i_4	i_4	i_1	i_1

a) Find the assignment when running the immediate acceptance algorithm (6 pts)

Step 1:

i_1 proposes to s_2

i_2 proposes to s_2

i_3 proposes to s_2

i_4 proposes to s_2

s_2 rejects i_2, i_3, i_4. i_1 is matched to s_2

Step 2:

i_2 proposes to s_1

i_3 proposes to s_3

i_4 proposes to s_3

s_3 rejects i_3. i_2 is matched to s_1, and i_4 is matched to s_3

Step 3:

i_3 proposes to s_4

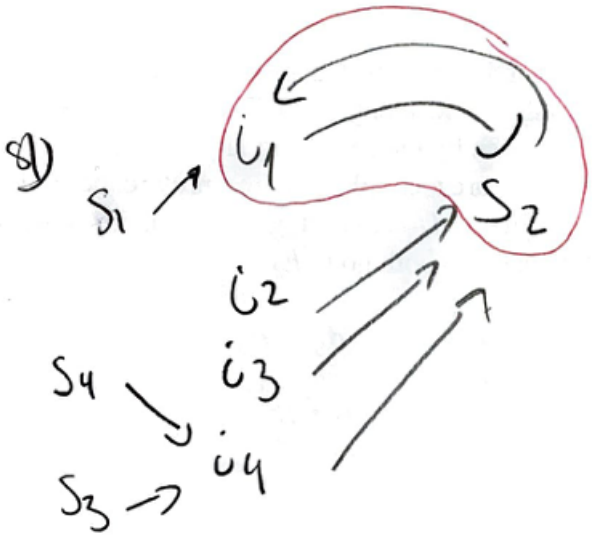
i_3 is matched to s_4

$$\mu(i_1) = s_2, \mu(i_2) = s_1, \mu(i_3) = s_4, \mu(i_4) = s_3$$

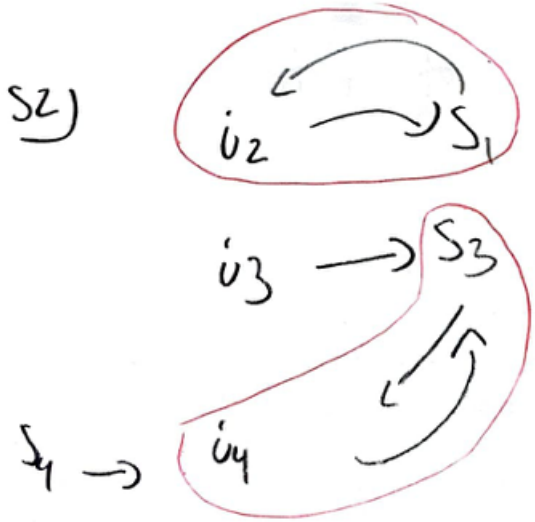
b) Would any student benefit from misrepresenting their preferences (6 pts)

Yes, for i_3 it would be better to list s_3 as their first preference. That way, they would get s_3 in the first step.

c) Find the assignment when running the top trading cycles algorithm (using the true preferences and priorities) (8 pts)

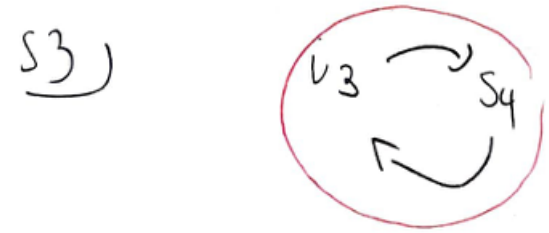


$\mu(i_1) = S_2$



$\mu(i_2) = S_1$

$\mu(i_4) = S_3$



$\mu(i_3) = S_4$

Problem 3 (20 points):

In a medical residence matching program there are 3 hospitals and 6 doctors. Hospital 1 can admit up to 3 doctors. Hospitals 2 and 3 are smaller and can only admit 2 residents each. The true preferences of doctors over hospitals and hospitals over doctors are given by the following tables (assume hospitals preferences are responsive). Note that doctors d_5 and d_6 are a couple and their preferences are expressed as pairs of hospitals: The first preference of d_5 & d_6 is to both go to h_1, their second preference is to both go to h_3, and their third preference is to both go to h_2.

$\boldsymbol{d_1}$	$\boldsymbol{d_2}$	$\boldsymbol{d_3}$	$\boldsymbol{d_4}$	$\boldsymbol{d_5}$ **&** $\boldsymbol{d_6}$
h_1	h_1	h_1	h_1	(h_1, h_1)
h_3	h_2	h_3	h_2	(h_3, h_3)
h_2	h_3	h_2	h_3	(h_2, h_2)

$\boldsymbol{h_1}$	$\boldsymbol{h_2}$	$\boldsymbol{h_3}$
d_5	d_5	d_6
d_4	d_3	d_2
d_1	d_1	d_3
d_2	d_2	d_1
d_3	d_4	d_4
d_6	d_6	d_5

a) Use the following algorithm to match hospitals and doctors (15 pts):

Step 1: Use the DA algorithm with doctors proposing excluding the couple (d_5 & d_6) from the problem

Step 2: Now manually match the couple to hospitals in order of their preferences (you must match them as a couple). When doing this manual matching, consider that the couple can displace doctors that are already matched as long as you leave hospital hiring them with a set of doctors that is preferred to the matchings they had obtained in step 1.

Step 3: Now manually match the doctor(s) displaced in step 2 to a hospital in order of their preferences. In this step unmatched doctors cannot displace doctors that have already been matched, so they can only be matched to hospitals that have a vacancy.

STEP 1:

We use DA without considering d_5 and d_6:

DA Step 1:

d_1 proposes to h_1

d_2 proposes to h_1

d_3 proposes to h_1

d_4 proposes to h_1

h_1 rejects d_3

DA Step 2:

d_3 propose to h_3

END: $\mu(h_1) = (d_1, d_2, d_4), \mu(h_3) = d_3$

STEP 2:

We try to manually match (d_5, d_6) to h_1. We can do this as long as we leave h_1 with a set of doctors it prefers. To compare set of doctors, we can need to use responsive preferences. We cannot compare (d_5, d_6, d_4) with (d_1, d_2, d_4) using responsive preferences. Thus, it is impossible to know if h_1 would be better off, and we cannot match (d_5, d_6) to h_1

Next, we try to manually match (d_5, d_6) to h_3. In this case, we know that $(d_5, d_6) P_H^{\#}(d_3, \emptyset)$, so the couple can displace d_3 and now $\mu(h_3) = (d_5, d_6)$

STEP 3:

We match d_3 to h_2. The final matching is: $\mu(h_1) = (d_1, d_2, d_4), \mu(h_2) = d_3, \mu(h_3) = (d_5, d_6)$.

b) For the doctor(s) displaced in step 3. Can you find a hospital that would be willing to form a blocking pair with them? (Note that you must treat the couple as an indivisible couple: If you want to remove one doctor of the couple from a hospital, you must remove both). (5 pts)

Only d_3 was displaced. This doctor prefers h_1 and h_3 to their current match. However, none of these would like to form a blocking pair because:

- d_3 is ranked lower than the doctors currently matched to h_1

- While h_3 like d_3 better than d_5, if h_3 changes d_3 for d_5 then d_6 will also leave (because they are couple with d_5, and we know that $(d_5, d_6) P_H^{\#}(d_3, \emptyset)$.

NOTE: I wrote "For the doctor(s) displaced in step 3" and I should have written "For the doctor(s) displaced is step 2" as no one is displaced in step 3. Thus, we will also consider correct the following answer: "There are no doctors displaced in step 3, so there will be no blocking pairs".